



 Houston, TX

 832.243.0700

 Houston Anodes

 Houston Anodes

 anodes@houstonanodes.com



HOUSTON ANODES

MANUFACTURER OF HIGH QUALITY ANODES



TABLE OF CONTENTS

A complete technical reference for aluminum and zinc sacrificial anode systems.

01	COMPANY About Houston Anodes · Safety & Employee Development · Quality Control & Laboratory · Electromagnetic Induction Technology	1
02	PLATFORM ANODES Overview · 90° Core Type A & B · 45° Core Type A & B · Straight Core Type A & B · Specifications · Alloy Chemistry · Electrochemical Properties	5
03	BRACELET ANODES Overview · Square Weld-on · Square Bolt-on · Tapered Weld-on & Bolt-on	21
04	PIER, PILING & BALLAST Overview · Trapezoidal Flush Core · Stand-off Core A & B · Rod Core · Rod Stand-off Core	36
05	TANKS & VESSELS Overview · Heater Treater & Eyebolt · Bowling Ball & Drum	62
06	ZINC HULLS Overview · Trapezoidal & Cast-in Core · Tapered & Tapered Trapezoidal · Teardrop	67
07	CUSTOM & REFERENCE Custom Anodes · Clamp, Mat & Link · Certifications & Standards · Contact Information	73

Standard sizes shown throughout this catalog. Any weight, size, shape, and configuration required by the customer can be manufactured. Weight based on aluminum alloy; also available in zinc alloy. When ordering, please specify metal and chemical composition.



PROTECTING OFFSHORE, SUBSEA AND ONSHORE INFRASTRUCTURE SINCE 1977

ANODES

Houston Anodes has manufactured sacrificial aluminum and zinc anodes for forty-seven years. Today, the company protects billions of dollars of infrastructure across five continents.

Every step of the process, from raw material testing and alloy formulation through casting, quality inspection, and delivery, takes place at the facility in the Energy Corridor of Houston, Texas.

The laboratory is accredited to ISO 17025. The manufacturing quality management system is certified ISO 9001:2015 by DNV. Electromagnetic induction technology homogenizes every alloy without flames and without contact.

50 YEARS

In business serving top operators worldwide

U.S. NAVY ANODES

Delivered with +35 year proven service life

ZERO CLAIMS

In 49 years





HOUSTON ANODES

SAFETY

EMPLOYEE SAFETY AND DEVELOPMENT

Houston Anodes operates on a holistic approach to workplace safety. The commitment is that no harm comes to employees or the environment. Prevention of all labor-related accidents, and continuous improvement of HSE performance, are embedded in how the business runs.



01 EMPLOYEE DEVELOPMENT

Ongoing technical and safety training at every level. Induction melting requires a more skilled operator than gas combustion, and the training reflects that.

02 SAFER TECHNOLOGY

Induction eliminates open flames and manual hand-stirring. Heat is generated by electromagnetic field without contact with the charge.

03 ZERO-HARM CULTURE

Prevention of all labor-related accidents is a core objective, not an aspiration. Active participation and support across the organization make it work.

04 HSE POLICY HIGHLIGHTS

- No open flames. Electromagnetic induction generates heat without fire.
- No manual stirring. Eddy currents homogenize the alloy automatically.
- Eliminates risk of superheating metals.
- Controlled temperature melting, adjusted per metal type.
- Continuous improvement through active stakeholder participation.
- Prevention of all labor-related accidents and environmental damage.



QUALITY CONTROL AND LABORATORY

The in-house laboratory is equipped with a Bruker-Quatron Magellan Q8 spectrometer, using the latest photomultiplier for analytical performance including analysis of non-metallic inclusions. It is configurable for virtually any metal analysis task.

- All raw materials inspected and tested upon arrival, prior to release into production.
- Chemical results verified against Mill Test Reports (MTRs).
- Chemical analysis performed on every heat.
- Electrochemical properties tested per DNV-RP-B401, NACETM-0190
- Rigorous inspection and testing points throughout production.
- Scheduled equipment calibration per OEM specifications.

ISO 17025

ACCREDITED LAB

ISO 9001

CERTIFIED BY DNV

NACE

TM-0190

100% USA

MANUFACTURING

HIGHEST PURITY RAW MATERIALS

Houston Anodes fabricates anodes using only raw materials of the highest purity, sourced from premium U.S. and global suppliers. Every delivery is inspected and tested in the ISO 17025-accredited laboratory before release into production.

99.9%

PRIMARY ALUMINUM

99.99%

SPECIAL HIGH GRADE ZINC

100% ASTM

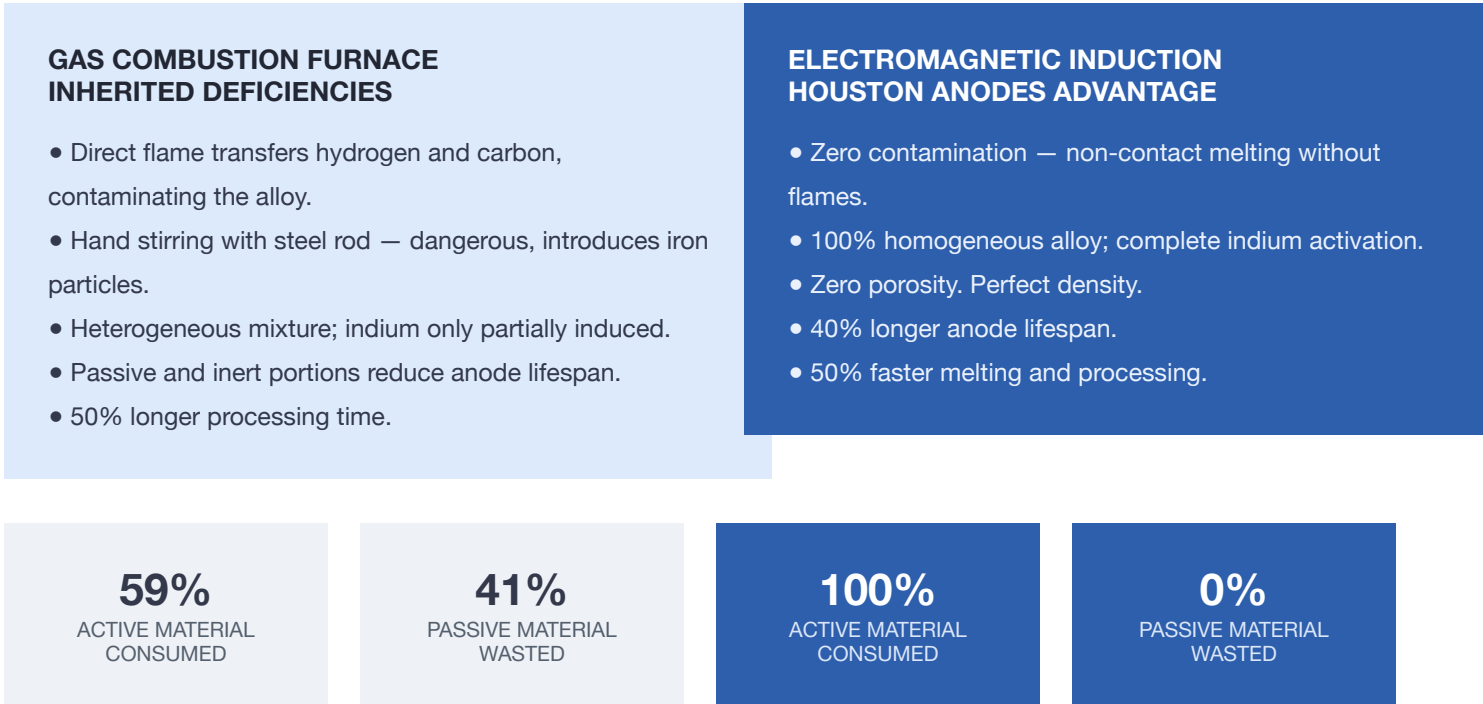
STEEL CORES

INDIUMNEVER MELT SCRAP OR
RECYCLED MATERIAL



ELECTROMAGNETIC INDUCTION TECHNOLOGY

The furnace is composed of a deionized water-cooled copper coil. When energized with an AC power source, it develops an electromagnetic field. Any metal charge kept inside the crucible is subjected to electromagnetic induction, creating eddy currents and an opposite circulating stirring flow of electrons within the molten mass. This generates heat in the charge depending on its resistivity — without flames and without touching the material. Gas combustion furnaces carry inherited deficiencies that compromise anode performance; electromagnetic induction removes them. The difference shows up in anode lifespan, processing time, and worker safety.



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PLATFORM ANODES

Houston Anodes aluminum platform and structural galvanic anodes are cast in low-iron, high-purity aluminum alloys. They can be manufactured in a wide variety of configurations for new construction or retrofitting.

All standard configurations are shown on the following pages. Houston Anodes can cast anodes with custom trapezoidal cross-sections to any required specifications.

01 CONFIGURATIONS

90° Core, Type A and Type B

45° Core, Type A and Type B

Straight Core, Type A and Type B

Weight Range: 180 to 1,250 lbs standard. Custom weights available.

Core Types: 2" SCH. 80, 3" SCH. 80, 4" SCH. 80

ALUMINUM
ALLOY

ZINC
ALLOY

CUSTOM CROSS-
SECTIONS





90° CORE TYPE A (1 OF 2)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF_180_AC90	60	93	6 ½	5 ½	180	225	2"SCH. 80
PF_285_AC90	120	153	5 ½	5 ½	285	361	2"SCH. 80
PF0_325_AC90	60	93	8 ¼	7 ¾	325	409	3"SCH. 80
PF1_325_AC90	96	129	7	5 ¾	325	455	3"SCH. 80
PF2_325_AC90	101	134	7	5 ¾	325	460	3"SCH. 80
PF3_325_AC90	120	153	5 ½	6 ¼	325	470	3"SCH. 80
PF_407_AC90	72	105	8	8 ¼	407	520	3"SCH. 80
PF0_411_AC90	72	105	8	8 ¼	411	520	3"SCH. 80
PF1_411_AC90	96	129	7	7	411	541	3"SCH.80
PF0_540_AC90	96	129	8	7 ¾	540	541	3"SCH. 80
PF1_540_AC90	98	131	6 ½	9 ¾	540	542	3"SCH. 80
PF0_725_AC90	95	128	9 ¾	9 ¾	725	905	4"SCH. 80
PF1_725_AC90	96	129	9 ¾	9 ¾	725	906	4"SCH. 80
PF2_725_AC90	97	130	9 ¾	9 ¾	725	907	4"SCH. 80

Standard sizes shown. Any weight and size required by the customer can be manufactured.



90° CORE TYPE A (2 OF 2)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF3_725_AC90	98	131	9 ¾	9 ¾	725	908	4"SCH. 80
PF_1050_AC90	144	164	9	9	1050	1255	4"SCH. 80
PF_1100_AC90	122	155	10 ½	10	1100	1316	4"SCH. 80
PF_1250_AC90	144	165	10	10 ½	1250	1455	4"SCH. 80

HOUSTON ANODES



90° CORE TYPE B (1 OF 2)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF_180_BC90	60	48	6 ½	5 ½	180	202	2"SCH. 80
PF_285_BC90	105	93	5 ½	5 ½	285	328	2"SCH. 80
PF0_325_BC90	60	48	8 ¼	7 ¾	325	368	3"SCH. 80
PF1_325_BC90	96	84	7	6	325	391	3"SCH. 80
PF2_325_BC90	101	89	7	5 ¾	325	394	3"SCH. 80
PF3_325_BC90	112	100	5 ½	6	325	400	3"SCH. 80
PF_407_BC90	72	60	8	8 ½	407	473	3"SCH. 80
PF0_411_BC90	72	60	8	8 ½	411	480	3"SCH. 80
PF1_411_BC90	96	84	7	7	411	480	3" SCH.80
PF0_540_BC90	96	84	8	7 ¾	540	606	3"SCH. 80
PF1_540_BC90	98	86	6 ½	9 ¾	540	607	3"SCH. 80
PF0_725_BC90	95	83	10	9 ½	725	813	4"SCH. 80
PF1_725_BC90	96	84	10	9	725	814	4"SCH. 80
PF2_725_BC90	97	85	9	10	725	814	4"SCH. 80



90° CORE TYPE B (2 OF 2)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF3_725_BC90	98	86	9	9 ¾	725	815	4"SCH. 80
PF_1050_BC90	144	120	9	9	1050	1200	4"SCH. 80
PF_1100_BC90	122	98	10 ½	10	1100	1120	4"SCH. 80
PF_1250_BC90	144	120	10	10 ½	1250	1400	4"SCH. 80

HOUSTON ANODES



45° CORE TYPE A (1 OF 2)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF_180_AC45	60	101	6 ½	5 ½	180	228	2"SCH. 80
PF_285_AC45	120	161	5 ½	5 ½	285	364	2"SCH. 80
PF0_325_AC45	60	101	8 ¼	7 ¾	325	414	3"SCH. 80
PF1_325_AC45	96	137	7	6	325	455	3"SCH. 80
PF2_325_AC45	101	142	7	5 ¾	325	460	3"SCH. 80
PF3_325_AC45	120	161	5 ½	6 ¼	325	470	3"SCH. 80
PF_407_AC45	72	113	8	8 ½	407	520	3"SCH. 80
PF0_411_AC45	72	113	8	8 ½	411	520	3"SCH. 80
PF1_411_AC45	96	137	7	7	411	541	3" SCH.80
PF0_540_AC45	96	137	8	7 ¾	540	541	3"SCH. 80
PF1_540_AC45	98	139	6 ½	9 ¾	540	542	3"SCH. 80
PF0_725_AC45	95	136	9 ¾	9 ¾	725	910	4"SCH. 80
PF1_725_AC45	96	137	9 ¾	9 ¾	725	911	4"SCH. 80
PF2_725_AC45	97	138	9 ¾	9 ¾	725	912	4"SCH. 80

Standard sizes shown. Any weight and size required by the customer can be manufactured.



45° CORE TYPE A (2 OF 2)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF3_725_AC45	98	139	9 ¾	9 ¾	725	913	4"SCH. 80
PF_1050_AC45	144	192	9	9	1050	1290	4"SCH. 80
PF_1100_AC45	122	163	10 ½	10	1100	1238	4"SCH. 80
PF_1250_AC45	144	192	10	10 ½	1250	1490	4"SCH. 80

HOUSTON ANODES

Standard sizes shown. Any weight and size required by the customer can be manufactured.



45° CORE TYPE B (1 OF 2)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF_180_BC45	60	72	6 ½	5 ½	180	212	2"SCH. 80
PF_285_BC45	105	117	5 ½	5 ½	285	328	2"SCH. 80
PF0_325_BC45	60	72	8 ¼	7 ¾	325	356	3"SCH. 80
PF1_325_BC45	96	108	7	6	325	378	3"SCH. 80
PF2_325_BC45	101	113	7	5 ¾	325	381	3"SCH. 80
PF3_325_BC45	112	124	5 ½	6	325	388	3"SCH. 80
PF_407_BC45	72	84	8	8 ½	407	445	3"SCH. 80
PF0_411_BC45	72	84	8	8 ½	411	464	3"SCH. 80
PF1_411_BC45	96	102	7	7	411	464	3" SCH.80
PF0_540_BC45	96	108	8	7 ¾	540	593	3"SCH. 80
PF1_540_BC45	98	110	6 ½	9 ¾	540	594	3"SCH. 80
PF0_725_BC45	95	107	10	9 ½	725	791	4"SCH. 80
PF1_725_BC45	96	108	10	9	725	791	4"SCH. 80
PF2_725_BC45	97	109	9	10	725	792	4"SCH. 80

Standard sizes shown. Any weight and size required by the customer can be manufactured.



45° CORE TYPE B (2 OF 2)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF3_725_BC45	98	110	9	9 ¾	725	793	4"SCH. 80
PF_1050_BC45	144	156	9	9	1050	1245	4"SCH. 80
PF_1100_BC45	122	134	10 ½	10	1100	1188	4"SCH. 80
PF_1250_BC45	144	156	10	10 ½	1250	1445	4"SCH. 80

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STRAIGHT CORE TYPE A (1 OF 2)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF_180_SAC	60	70	6 ½	5 ½	180	210	2"SCH. 80
PF_285_SAC	120	132	5 ½	5 ½	285	340	2"SCH. 80
PF0_325_SAC	60	70	8 ¼	7 ¾	325	385	3"SCH. 80
PF1_325_SAC	96	106	7	6	325	416	3"SCH. 80
PF2_325_SAC	101	111	7	5 ¾	325	420	3"SCH. 80
PF3_325_SAC	120	132	5 ½	6 ¼	325	438	3"SCH. 80
PF_407_SAC	72	82	8	8 ½	407	477	3"SCH. 80
PF0_411_SAC	72	82	8	8 ½	411	481	3"SCH. 80
PF1_411_SAC	96	106	7	7	411	502	3" SCH.80
PF0_540_SAC	96	106	8	7 ¾	540	631	3"SCH. 80
PF1_540_SAC	98	108	6 ½	9 ¾	540	632	3"SCH. 80
PF0_725_SAC	95	105	9 ¾	9 ¾	725	856	4"SCH. 80
PF1_725_SAC	96	106	9 ¾	9 ¾	725	857	4"SCH. 80
PF2_725_SAC	97	107	9 ¾	9 ½	725	858	4"SCH. 80

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STRAIGHT CORE TYPE A (2 OF 2)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF3_725_SAC	98	108	9 ¾	9 ¾	725	859	4"SCH. 80
PF_1050_SAC	144	168	9	9	1050	1260	4"SCH. 80
PF_1100_SAC	122	132	10 ½	10	1100	1265	4"SCH. 80
PF_1250_SAC	144	168	10	10 ½	1250	1460	4"SCH. 80

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STRAIGHT CORE TYPE B (1 OF 2)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF_180_SBC	60	30	6 ½	5 ½	180	196	2"SCH. 80
PF_285_SBC	120	90	5 ½	5 ½	285	340	2"SCH. 80
PF0_325_SBC	60	30	8 ¼	7 ¾	325	374	3"SCH. 80
PF1_325_SBC	96	66	7	6	325	397	3"SCH. 80
PF2_325_SBC	101	71	7	5 ¾	325	400	3"SCH. 80
PF3_325_SBC	120	90	5 ½	6 ¼	325	411	3"SCH. 80
PF_407_SBC	72	42	8	8 ½	407	480	3"SCH. 80
PF0_411_SBC	72	42	8	8 ½	411	484	3"SCH. 80
PF1_411_SBC	96	66	7	7	411	484	3"SCH.80
PF0_540_SBC	96	66	8	7 ¾	540	612	3"SCH. 80
PF1_540_SBC	98	68	6 ½	9 ¾	540	613	3"SCH. 80
PF0_725_SBC	95	65	9 ¾	9 ¾	725	878	4"SCH. 80
PF1_725_SBC	96	66	9 ¾	9 ¾	725	879	4"SCH. 80
PF2_725_SBC	97	67	9 ¾	9 ½	725	880	4"SCH. 80

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STRAIGHT CORE TYPE B (2 OF 2)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
PF3_725_SBC	98	68	9 ¾	9 ¾	725	881	4"SCH. 80
PF_1050_SBC	144	114	9	9	1050	1187	4"SCH. 80
PF_1100_SBC	122	92	10 ½	10	1100	1218	4"SCH. 80
PF_1250_SBC	144	114	10	10 ½	1250	1387	4"SCH. 80

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SPECIFICATIONS

Metric dimensional reference — straight, 90° and 45° core configurations.

Anode Type (Straight Core)	Anode Type (90° Core)	Anode Type (45° Core)	Gross Weight (kg)	Net Weight (kg)	L (mm)	W1 (mm)	W2 (mm)	T (mm)	Lc (mm)
A59.0S1	-BC90	-BC45	68.3	59	750	160	220	169	1170
A148.0S1	-BC90	-BC45	164.5	148	1600	180	225	184	2600
A320.0S1	-BC90	-BC45	389	320	2500	220	265	238	3100
A356.0S1	-BC90	-BC45	423.1	356	2440	230	290	247	3050
A533.0S1	-BC90	-BC45	603	533	2570	310	335	270	4600

Consult a CP specialist for guidance at elevated temperatures — performance de-rating will be needed. Exercise care with variable ballast tanks: dangerous volumes of hydrogen can be generated under certain conditions.

HOUSTON ANODES



ALLOY CHEMISTRY

Two premium low-iron aluminum alloys, engineered for deep water and shallow water cathodic protection service.

HA-DW

DEEP WATER ALLOY

Iron (Fe)	0.07 max
Silicon (Si)	0.10 max
Copper (Cu)	0.003 max
Zinc (Zn)	4.75 – 5.25
Indium (In)	0.015 – 0.025
Titanium (Ti)	0.025 max
Cadmium (Cd)	Part of others
Others (each)	0.02 max
Aluminum (Al)	Remainder

HA-SW

SHALLOW WATER ALLOY

Iron (Fe)	0.10 max
Silicon (Si)	0.10 max
Copper (Cu)	0.006 max
Zinc (Zn)	3.00 – 6.00
Indium (In)	0.010 – 0.020
Titanium (Ti)	0.025 max
Cadmium (Cd)	Part of others
Others (each)	0.02 max
Aluminum (Al)	Remainder

Values shown in % by weight. When ordering, please specify metal and chemical composition.



ELECTROCHEMICAL & PHYSICAL PROPERTIES

Design-basis potentials and current capacities for the HA-DW and HA-SW aluminum alloys.

(-) 1.08 V

OPEN CIRCUIT POTENTIAL
vs Ag/AgCl seawater

(-) 1.05 V

CLOSED CIRCUIT POTENTIAL¹
vs Ag/AgCl seawater

PROPERTY	HA-DW	HA-SW
Seawater capacity @ 25°C	1,100 A-hr/lb (2,420 A-hr/kg)	1,100 A-hr/lb (2,420 A-hr/kg)
Seawater capacity @ 5°C	1,100 A-hr/lb (2,420 A-hr/kg)	Variable ²
Seabed mud capacity ³	950 A-hr/lb (2,090 A-hr/kg)	950 A-hr/lb (2,090 A-hr/kg)
Seabed mud capacity @ 5°C	950 A-hr/lb (2,090 A-hr/kg)	Variable ²
Alloy density	0.098 lb/in ³ (2.71 g/cm ³)	0.098 lb/in ³ (2.71 g/cm ³)

RECOMMENDED ENVIRONMENT

- HA-DW — Any ocean, any depth, any temperature.
- HA-SW — Tropical ocean < 600 fsw (180 msw), mean water temp. > 45°F (7°C).

NOTES

1. Recommended value for CP system design.
2. Activation can be a problem with chemistries at extremes of specification in cold water.
3. Recommended value for CP system design of offshore pipelines using bracelet or sub-bottom anode systems.
- Consult a CP specialist for guidance at elevated temperatures — performance de-rating will be needed.
- Exercise care with variable ballast tanks: dangerous volumes of hydrogen can be generated under certain conditions.

BRACELET ANODES

Houston Anodes aluminum bracelet anodes are cast in low-iron, high-purity aluminum alloys. They can be manufactured in a wide variety of diameters using an extensive inventory of water-cooled molds.

01 CONFIGURATIONS

Square Weld-on

Square Bolt-on

Tapered Weld-on

Tapered Bolt-on

Diameter Range: 4½" to 48" standard. Custom diameters available.

Weight Range: 22 to 265 lbs standard. Custom weights available.

ALUMINUM ALLOY

ZINC ALLOY

CUSTOM CROSS-SECTIONS



HOUSTON
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SQUARE WELD-ON (1 OF 5)



NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR4.5_24_WO	4 ½	10 ½	1 ¼	1 ½	24	27	3/16 x 1
BR4.5_26_WO	4 ½	12	1 ¼	1 ½	26	29	3/16 x 1
BR4.5_29_WO	4 ½	12	1 ¼	1 ½	29	32	3/16 x 1
BR6.625_37_WO	6 ½	11 ¼	1 ½	1 ½	37	46	3/16 x 1 ½
BR6.625_43_WO	6 ½	16 ½	1	1 ½	43	52	3/16 x 1 ½
BR8_73_WO	8	13 ¾	2	1 ½	73	82	3/16 x 1 ½
BR8_74_WO	8	13 ½	1	2	74	83	3/16 x 1 ½
BR8.625_50_WO	8 ½	13	2	1 ½	50	59	3/16 x 1 ½
BR8.625_52_WO	8 ½	13 ½	2	1 ½	52	61	3/16 x 1 ½
BR8.625_57_WO	8 ½	13 ½	1 ½	1 ½	57	66	3/16 x 1 ½
BR8.625_80_WO	8 ½	13 ½	1 ½	2	80	91	3/16 x 1 ½
BR8.625_82_WO	8 ½	13 ¾	2	2	82	91	3/16 x 1 ½
BR8.625_118_WO	8 5/8	16	1 ½	2 ½	118	122	3/16 x 1 ½
BR8.625_124_WO	8 ½	16	1 ½	2 ½	124	133	3/16 x 1 ½



SQUARE WELD-ON (2 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR9_44_WO	9	11 ¼	3	1 ½	44	55	3/16 x 1 ½
BR9.5_84_WO	9 ½	10 ½	2	2 ½	84	95	3/16 x 1 ½
BR10_95_WO	10	11	1 ½	2 ½	95	106	3/16 x 1 ½
BR10.5_133_WO	10 ½	22 ½	1 ½	1 ¼	133	144	3/16 x 1 ½
BR10.625_49_WO	10 ⅝	15 ½	2	1	49	60	3/16 x 1 ½
BR10.75_35_WO	10 ¾	10 ¾	1 ½	1	35	46	3/16 x 1 ½
BR11_58_WO	11	18	1 ½	1	58	69	3/16 x 1 ½
BR11_240_WO	11	36	1 ½	2 ½	240	255	¼ x 2
BR12_80_WO	12	22 ½	2	1	80	94	¼ x 2
BR12.75_88_WO	12 ¾	23 ¾	1 ½	1	88	102	¼ x 2
BR12.75_111_WO	12 ¾	23	1 ½	1 ½	111	125	¼ x 2
BR13_143_WO	13	17	2	2	143	157	¼ x 2
BR14_88_WO	14	13 ¾	2	1 ½	88	102	¼ x 2
BR14_89_WO	14	13 ¾	1	1 ½	89	103	¼ x 2
BR14_115_WO	14	11	1 ½	2 ½	115	129	¼ x 2
BR14_96.5_WO	14	12 ¾	1 ½	2	96.5	110	¼ x 2
BR16_116_WO	16	16	1 ½	1 ½	116	132	¼ x 2
BR16_145_WO	16	16 ¼	2	1 ¾	145	161	¼ x 2
BR16_231_WO	16	25 ½	1 ½	2 ½	231	247	¼ x 2
BR16_103_WO	16	16 ¼	2	1 ½	103	119	¼ x 2
BR16_160_WO	16	26 ¼	1 ½	1 ¼	160	176	¼ x 2



SQUARE WELD-ON (3 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR16_166_WO	16	11 ½	2	2 ¾	166	182	¼ x 2
BR16_170_WO	16	25	1 ½	1 ¼	170	186	¼ x 2
BR18_265_WO	16	22 ½	2 ½	4 ½	265	281	¼ x 2
BR18_64_WO	18	12	2	1	64	81	¼ x 2
BR18_77_WO	18	9 ¾	1	1 ½	77	94	¼ x 2
BR18_81_WO	18	11 ¾	1	1 ½	81	98	¼ x 2
BR18_83_WO	18	9 ¾	1	1 ½	83	100	¼ x 2
BR18_98_WO	18	10	1 ½	1 ¾	98	114	¼ x 2
BR18_101_WO	18	10	1 ½	1 ¾	101	115	¼ x 2
BR19.5_172_WO	19 ½	19	1 ½	1 ½	172	189	¼ x 2
BR20_108_WO	20	14	2	1 ½	108	128	¼ x 2
BR20_130_WO	20	16 ¼	1 ½	3	130	150	¼ x 2
BR20_132_WO	20	14 ½	1 ½	2 ½	132	152	¼ x 2
BR20_134_WO	20	14 ¼	2 ¼	1 ½	134	154	¼ x 2
BR20_151_WO	20	12	1 ½	2	151	171	¼ x 2
BR20_152_WO	20	12	1 ½	2	152	168	¼ x 2
BR20_169_WO	20	22 ¼	2 ½	1 ¼	169	189	¼ x 2
BR20_169b_WO	20	15 ½	2 ½	1 ¾	169	189	¼ x 2
BR20_168_WO	20	18 ½	3	1 ½	168	188	¼ x 2
BR20_168b_WO	20	22 ¼	3	1 ¾	168	188	¼ x 2
BR20_171_WO	20	15 ½	3	1 ¾	171	191	¼ x 2



SQUARE WELD-ON (4 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR20_177_WO	20	14	2 ½	2	177	197	¼ x 2
BR20_189_WO	20	19 ½	3	1 ½	189	209	¼ x 2
BR20_192_WO	20	15	2 ½	1 ¾	192	212	¼ x 2
BR20_192b_WO	20	18	3	2 ¾	192	212	¼ x 2
BR20_207_WO	20	13 ¾	3	1 ¾	207	227	¼ x 2
BR20_226_WO	20	16 ¾	2	2	226	246	¼ x 2
BR20_227_WO	20	18	3	2	227	247	¼ x 2
BR20_258_WO	20	18	3	1 ½	258	278	¼ x 2
BR20_260_WO	20	13 ¾	3	1 ½	260	280	¼ x 2
BR22.625_143_WO	22 ½	20	1 ½	2 ½	143	175	¼ x 2
BR24_169_WO	24	13	2 ½	1 ¾	169	194	¼ x 2
BR24_185_WO	24	6 ½	2 ½	3 ½	185	210	¼ x 2
BR24_188_WO	24	6 ½	3	3 ½	188	213	¼ x 2
BR24_188b_WO	24	6 ¾	3	3 ½	188	213	¼ x 2
BR24_189_WO	24	9	3	2 ¾	189	214	¼ x 2
BR24_193_WO	24	7	2 ½	3 ½	193	218	¼ x 2
BR24_195_WO	24	18	2 ½	1 ½	195	220	¼ x 2
BR24_197_WO	24	13 ¾	3	2	197	222	¼ x 2
BR24_159_WO	24	10 ½	3	2	159	184	¼ x 2
BR24_220_WO	24	20	2	1 ½	220	245	¼ x 2
BR24_268_WO	24	12 ½	1 ½	3 ¾	268	293	¼ x 2



SQUARE WELD-ON (5 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR24_332_WO	24	13 ¾	3	4	332	357	¼ x 2
BR24_335_WO	24	13 ¾	2 ½	3 ¾	335	360	¼ x 2
BR24_497_WO	24	15 ½	3	4	497	522	¼ x 2
BR30_164_WO	30	12	1 ½	2	164	192	¼ x 2
BR30_220_WO	30	12	3 ½	2	220	248	¼ x 2
BR30_220b_WO	30	12	2 ½	2	220	248	¼ x 2
BR30_257_WO	30	14	3	2	257	285	¼ x 2
BR30_284_WO	30	15 ½	3 ½	2	284	312	¼ x 2

HOUSTON ANODES



SQUARE BOLT-ON (1 OF 5)



NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR4.5_22_BO	4 ½	10 ½	1 ¼	1 ¼	22	25	3/16 x 1
BR4.5_24_BO	4 ½	12	1 ¼	1 ¼	24	27	3/16 x 1
BR4.5_27_BO	4 ½	12	1 ¼	1 ¼	27	30	3/16 x 1
BR6.625_35_BO	6 ½	11 ¼	1 ½	1 ¼	35	44	3/16 x 1 ½
BR6.625_41_BO	6 ½	16 ½	1	1 ¼	41	52	3/16 x 1 ½
BR8_71_BO	8	13 ¾	2	1 ½	70	159	3/16 x 1 ½
BR8_72_BO	8	13 ½	1	2	72	61	3/16 x 1 ½
BR8.625_48_BO	8 ½	13	2	1 ½	48	46	3/16 x 1 ½
BR8.625_50_BO	8 ½	13 ½	2	1 ½	50	39	3/16 x 1 ½
BR8.625_55_BO	8 ½	13 ½	1 ½	1 ½	55	64	3/16 x 1 ½
BR8.625_78_BO	8 ½	13 ¼	1 ½	2	78	87	3/16 x 1 ½
BR8.625_80_BO	8 ½	13 ¾	1 ½	2	80	89	3/16 x 1 ½
BR8.625_122_BO	8 ½	16	1 ½	2 ½	122	129	3/16 x 1 ½
BR9_42_BO	9	11 ¼	3	1 ½	42	51	3/16 x 1 ½



SQUARE BOLT-ON (2 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR9.5_82_BO	9 ½	10 ¼	2	2 ½	82	91	3/16 x 1 ½
BR10_93_BO	10	11	1 ½	2 ½	93	104	3/16 x 1 ½
BR10.5_131_BO	10 ½	22 ½	1 ½	1 ¼	131	142	3/16 x 1 ½
BR10.625_47_BO	10 ⅝	15 ½	2	1	47	58	3/16 x 1 ½
BR10.75_33_BO	10 ¾	10 ¾	2	1	33	44	3/16 x 1 ½
BR11_56_BO	11	18	1 ½	1	56	67	3/16 x 1 ½
BR11_238_BO	11	36	1 ½	2 ½	238	253	¼ x 2
BR12_78_BO	12	22 ½	1	1 ½	78	92	¼ x 2
BR12.75_86_BO	12 ¾	23 ¾	1 ½	1	86	102	¼ x 2
BR12.75_109_BO	12 ¾	23	1 ½	1 ¼	109	123	¼ x 2
BR13_141_BO	13	17	2	2	141	154	¼ x 2
BR14_86_BO	14	13 ¾	2	1 ½	86	99	¼ x 2
BR14_87_BO	14	13 ¾	1	1 ½	87	100	¼ x 2
BR14_94.5_BO	14	12 ¾	1	2	94.5	108	¼ x 2
BR14_113_BO	14	11	1 ½	2 ½	113	126	¼ x 2
BR16_114_BO	16	16	2	1 ½	114	127	¼ x 2
BR16_143_BO	16	16 ¼	2	1 ¾	143	156	¼ x 2
BR16_229_BO	16	25 ½	1 ½	1 ¼	229	246	¼ x 2
BR16_101_BO	16	16 ¼	1 ½	1 ¼	101	116	¼ x 2
BR16_158_BO	16	26 ¼	1 ½	1 ¼	158	171	¼ x 2
BR16_164_BO	16	11 ½	2	2 ½	164	177	¼ x 2



SQUARE BOLT-ON (3 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR16_168_BO	16	25	1 ½	1 ¼	168	181	¼ x 2
BR16_263_BO	16	22 ½	2 ½	4 ½	263	279	¼ x 2
BR18_62_BO	18	12	2	1	62	79	¼ x 2
BR18_75_BO	18	9 ¾	2	1 ½	75	92	¼ x 2
BR18_79_BO	18	11 ¾	1	1 ¼	79	96	¼ x 2
BR18_81_BO	18	9 ¾	1	1 ¼	81	98	¼ x 2
BR18_96_BO	18	10	1 ½	1 ¼	96	112	¼ x 2
BR18_99_BO	18	10	1 ½	1 ¼	99	115	¼ x 2
BR19.5_170_BO	19 ½	19	1 ½	1	170	187	¼ x 2
BR20_106_BO	20	14	2	1 ½	106	130	¼ x 2
BR20_128_BO	20	16 ¼	1 ½	3	128	146	¼ x 2
BR20_130_BO	20	14 ½	1 ½	2 ½	130	148	¼ x 2
BR20_132_BO	20	14 ¼	2 ½	1	132	153	¼ x 2
BR20_149_BO	20	12	1 ½	2	149	169	¼ x 2
BR20_150_BO	20	12	1 ½	2	150	166	¼ x 2
BR20_167_BO	20	22 ¼	2 ½	1 ¼	167	187	¼ x 2
BR20_167b_BO	20	15 ½	2 ½	1 ¾	167	187	¼ x 2
BR20_166_BO	20	18 ½	3	1 ½	166	186	¼ x 2
BR20_166b_BO	20	22 ¼	2 ½	1 ¼	166	186	¼ x 2
BR20_164_BO	20	15 ½	3	1 ¼	164	184	¼ x 2
BR20_169_BO	20	14	2 ½	2	169	189	¼ x 2



SQUARE BOLT-ON (4 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR20_175_BO	20	19 ½	3	1 ½	175	195	¼ x 2
BR20_187_BO	20	15	2 ½	1 ¾	187	207	¼ x 2
BR20_190_BO	20	18	3	1 ¾	190	210	¼ x 2
BR20_190b_BO	20	13 ¾	3	2 ¾	190	210	¼ x 2
BR20_205_BO	20	16 ¾	2	2	205	225	¼ x 2
BR20_224_BO	20	18	2	2	224	244	¼ x 2
BR20_225_BO	20	18	3	2	225	295	¼ x 2
BR20_256_BO	20	13 ¾	3	3	256	276	¼ x 2
BR20_258_BO	20	13 ¾	3	3	258	278	¼ x 2
BR22.625_141_BO	22 ½	20	1 ½	2 ½	141	173	¼ x 2
BR24_167_BO	24	13	2 ½	1 ¾	167	192	¼ x 2
BR24_183_BO	24	6 ½	2 ½	3 ½	183	208	¼ x 2
BR24_186_BO	24	6 ½	2 ½	3 ½	186	211	¼ x 2
BR24_186b_BO	24	6 ½	3	3 ½	186	211	¼ x 2
BR24_187_BO	24	9	3	2 ¾	187	212	¼ x 2
BR24_191_BO	24	7	2 ½	3 ½	191	216	¼ x 2
BR24_193_BO	24	18	2 ½	1 ½	193	218	¼ x 2
BR24_195_BO	24	13 ¾	3	2	195	220	¼ x 2
BR24_157_BO	24	10 ½	3	2	157	182	¼ x 2
BR24_218_BO	24	20	2	1 ½	218	243	¼ x 2
BR24_266_BO	24	12 ½	1 ½	3 ½	266	291	¼ x 2



SQUARE BOLT-ON (5 OF 5)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BR24_330_BO	24	10 ½	3	4	330	255	¼ x 2
BR24_333_BO	24	13 ¾	2 ½	3 ½	333	258	¼ x 2
BR24_495_BO	24	15 ½	3	4	495	520	¼ x 2
BR30_162_BO	30	12	3	1 ½	162	190	¼ x 2
BR30_118_BO	30	12	3 ½	2	118	246	¼ x 2
BR30_255_BO	30	12	2 ½	2	255	283	¼ x 2
BR30_282_BO	30	14	3	2	282	310	¼ x 2
BR30_290_BO	30	15 ½	3 ½	2	290	356	¼ x 2
BR30_328_BO	30	21 ¼	4	2 ½	228	356	¼ x 2
BR30_333_BO	30	16	3	2	333	361	¼ x 2

HOUSTON ANODES



TAPERED WELD-ON (1 OF 2)



NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BRT2.375_13_WO	2 ½	12	1	1 ¼	13	15	3/16 x 1
BRT3.5_17_WO	3 ½	12	1	1 ¼	17	20	3/16 x 1
BRT4.5_26_WO	4 ½	12 ½	1 ¼	1 ¼	26	31	3/16 x 1
BRT5.563_50_WO	5 ½	13 ½	1	2	50	54	3/16 x 1
BRT4.5_41_WO	4 ½	12 ½	1 ¼	2 ½	41	45	3/16 x 1
BRT6.625_35_WO	6 ½	11 ¼	1 ½	1 ½	35	42	3/16 x 1 ½
BRT8.625_30_WO	8 ½	8 ½	2	1 ½	30	38	3/16 x 1 ½
BRT8.625_45_WO	8 ½	12	1 ½	1 ½	45	53	3/16 x 1 ½
BRT8.625_79_WO	8 ½	16	2 ½	2	79	87	3/16 x 1 ½
BRT8.625_82_WO	8 ½	19 ½	1 ½	1 ½	82	90	3/16 x 1 ½
BRT8.625_85_WO	8 ½	16	1 ½	2	85	93	3/16 x 1 ½
BRT10_37_WO	10	8 ½	2	1 ½	37	41	3/16 x 1 ½
BRT12.75_149_WO	12 ½	20 ¼	2	2	149	161	3/16 x 1 ½
BRT13_155_WO	13	21	2	2	155	167	3/16 x 1 ½

Standard sizes shown. Any weight and size required by the customer can be manufactured.



TAPERED WELD-ON (2 OF 2)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BRT20_202_WO	20	13 ½	3	3	202	227	¼ x 2
BRT32.25_328_WO	32 ¼	15 ½	2 ½	2 ½	328	354	¼ x 2
BRT36.25_382_WO	36 ¼	17 ½	3	2 ½	382	410	¼ x 2

HOUSTON ANODES



TAPERED BOLT-ON (1 OF 2)



NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BRT2.375_11_BO	2 ½	12	1	1 ¼	11	13	3/16 x 1
BRT3.5_15_BO	3 ½	12	1	1 ¼	15	18	3/16 x 1
BRT4.5_23_BO	4 ½	12 ½	1 ¼	1 ¼	23	26	3/16 x 1
BRT4.5_39_BO	4 ½	12 ½	1 ¼	2 ½	39	42	3/16 x 1
BRT5.563_50_BO	5 ½	13 ½	1	2	50	54	3/16 x 1
BRT6.625_31_BO	6 ½	11 ¼	1 ½	1 ½	31	40	3/16 x 1 ½
BRT8.625_39_BO	8 ½	12	1 ½	1 ½	39	50	3/16 x 1 ½
BRT8.625_46_BO	8 ½	13	1 ½	1 ½	46	57	3/16 x 1 ½
BRT8.625_82_BO	8 ½	19 ¼	1 ½	1 ½	82	90	3/16 x 1 ½
BRT8.625_88_BO	8 ½	16	1 ½	2	88	99	3/16 x 1 ½
BRT8.875_97_BO	8 ½	16 ½	1 ½	2 ½	97	108	3/16 x 1 ½
BRT10_37_BO	10	8 ¾	2	1 ½	37	41	3/16 x 1 ½
BRT10.625_74_BO	10 ½	11	1 ½	2 ½	74	85	3/16 x 1 ½
BRT12.75_149_BO	12 ¾	20 ¼	2	2	149	161	3/16 x 1 ½



TAPERED BOLT-ON (2 OF 2)

NAME	Ø (in)	L (in)	G (in)	T (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BRT13_155_BO	13	21	2	2	155	167	3/16 x 1 ½

HOUSTON ANODES

PIER, PILING & BALLAST

Houston Anodes aluminum galvanic anodes for piers and pilings are cast in one of two premium aluminum alloys for optimal performance in any conditions. Core length comes standard depending on size, and holes may be drilled for bolt-on applications.

01 CONFIGURATIONS

Trapezoidal Flush Core

Trapezoidal Stand-off Core Type A

Trapezoidal Stand-off Core Type B

Trapezoidal Rod Core

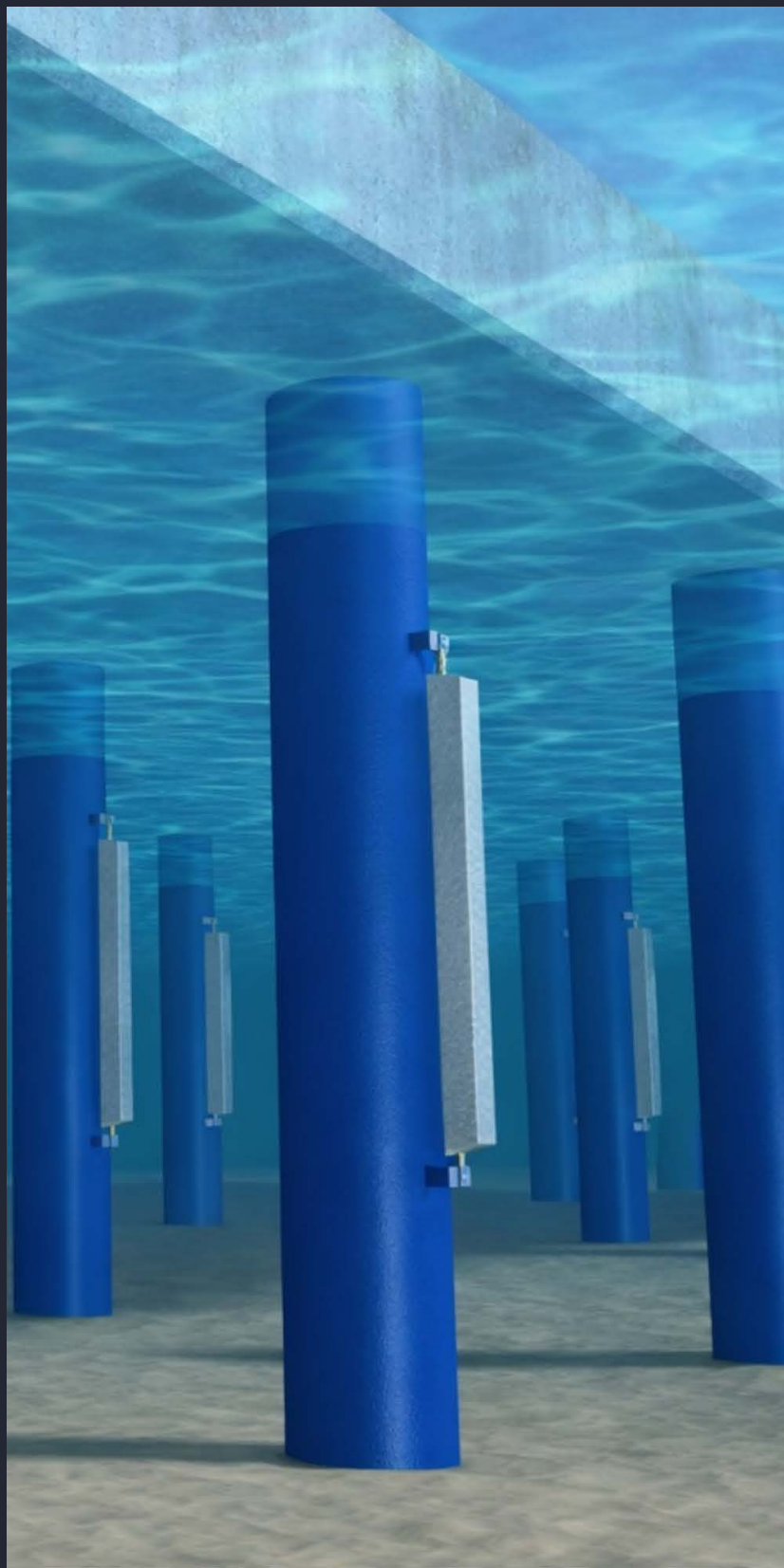
Trapezoidal Rod Stand-off Core

ALUMINUM ALLOY

ZINC ALLOY

CUSTOM CROSS-SECTIONS

HOUSTON ANODES





FLUSH CORE (1 OF 5)



NAME	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_0.5_FC	2 ½	1 ½	1 ½	0.5	0.8
TR_0.85_FC	4	1 ½	1 ½	0.85	1.1
TR_1_FC	5	1 ½	1 ½	1	1.6
TR_2_FC	10	1 ½	1 ½	2	2.9
TR0_2.2_FC	6	2	2	2.2	2.8
TR1_2.2_FC	6 ½	3	1 ½	2.2	2.9
TR0_3_FC	11	1 ½	2	3	3.9
TR1_3_FC	4	4	2	3	3.5
TR2_3_FC	8	2	2	3	3.7
TR_3.8_FC	8	4	1 ½	3.8	4.5
TR_4.1_FC	12	3	1½	4.1	5.1
TR0_5_FC	9	3	2	5	8
TR1_5_FC	24	1 ½	1 ½	5	6.7
TR_5.7_FC	30	1 ½	1	5.7	7.8



FLUSH CORE (2 OF 5)

NAME	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_5.8_FC	5 ½	5 ½	2	5.8	6.4
TR_6.3_FC	6	5 ½	2	6.3	6.9
TR_6.5_FC	22	1 ½	2	6.5	8.1
TR_7.7_FC	9	4 ½	2	7.7	8.5
TR0_8_FC	12	2	3 ½	8	9
TR1_8_FC	8	3	3 ½	8	8.7
TR_8.7_FC	8	3	3 ½	8.7	9.4
TR_8.8_FC	24	2	2	8.8	10.5
TR0_10_FC	10	3 ½	3	10	10.9
TR1_10_FC	48	1 ½	1 ½	10	10.9
TR_10.5_FC	9	3 ¾	3 ¾	10.5	11.3
TR_11_FC	17	2	3 ½	11	12
TR0_12_FC	12	3 ½	3	12	13
TR1_12_FC	60	1 ½	1 ½	12	16
TR_13_FC	10	5 ½	2 ½	13	14
TR_14_FC	18	4	2	14	15
TR_16_FC	23	3	2 ½	16	18
TR_17_FC	20 ½	4 ½	2	17	19
TR0_18_FC	48	2	2	18	21
TR1_18_FC	13 ½	5 ½	2 ½	18	19
TR2_18_FC	21 ½	3 ½	3	18	20

Standard sizes shown. Any weight and size required by the customer can be manufactured.



FLUSH CORE (3 OF 5)

NAME	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_20_FC	30	2	3 ½	20	22
TR0_22_FC	20	4 ½	2 ½	22	23
TR1_22_FC	60	2	2	22	26
TR_25_FC	16	4	4	25	26
TR0_26_FC	24	3	3 ½	26	27
TR1_26_FC	24	3 ½	3 ½	26	28
TR0_28_FC	48	2 ½	2 ½	28	31
TR1_28_FC	48	2 ½	2 ½	28	31
TR_32_FC	24	5 ½	2 ¾	32	34
TR_34_FC	22	4	4	34	35
TR_36_FC	60	2 ½	2 ½	36	39
TR0_42_FC	48	3	3	42	45
TR1_42_FC	36	5	2 ½	42	45
TR2_42_FC	36	3 ½	3 ½	42	45
TR_47_FC	55	3	3	47	51
TR0_48_FC	20	4 ½	5 ½	48	49
TR1_48_FC	20	5	5	48	50
TR_50_FC	60	2 ½	3 ½	50	54
TR0_52_FC	60	3	3	52	54
TR1_52_FC	22	5	5	52	54
TR_58_FC	60	2 ½	4	58	60

Standard sizes shown. Any weight and size required by the customer can be manufactured.



FLUSH CORE (4 OF 5)

NAME	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_59_FC	30	4 ½	4 ½	59	61
TR_65_FC	42	4	4	65	67
TR_72_FC	23	6 ½	5	72	75
TR_74_FC	48	4	4	74	77
TR_76_FC	44	4	4 ½	76	80
TR_86_FC	28	6	5 ½	86	88
TR_87_FC	36	5	5	87	90
TR_92_FC	60	4	4	92	96
TR_93_FC	48	5	4	93	96
TR_95_FC	36	6	4 ½	95	97
TR_97_FC	45 ½	4 ½	5	97	100
TR_103_FC	45	4 ½	5 ½	103	106
TR_105_FC	60 ½	4	4	105	108
TR_125_FC	40	6	5 ½	125	138
TR_130_FC	42	6	5 ½	130	133
TR_134_FC	77	4 ½	4	134	138
TR_138_FC	40	6 ½	5 ½	138	142
TR_140_FC	40	6 ½	5 ½	140	144
TR_148_FC	42 ½	6 ½	5 ½	148	152
TR_150_FC	77	4 ½	4 ½	150	155
TR_152_FC	40 ½	6 ½	6	152	156

Standard sizes shown. Any weight and size required by the customer can be manufactured.



FLUSH CORE (5 OF 5)

NAME	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_160_FC	60	5	5 ½	160	165
TR0_165_FC	36	7	7	165	168
TR1_165_FC	48	10 ½	3 ½	165	168
TR_174_FC	48	10	3 ½	174	178
TR_182_FC	72	4 ½	5 ½	182	186
TR_185_FC	60 ½	5 ½	6	185	189
TR_195_FC	61	5 ½	6	195	199
TR_231_FC	42 ½	7	8	231	236
TR_243_FC	51	7	7	243	248
TR_244_FC	51 ¼	7	7	244	249
TR_245_FC	51 ½	7	7	245	250
TR_258_FC	28	10 ½	9	258	263
TR_266_FC	49	7	8	266	271
TR_304_FC	48	6 ½	10	304	310



STAND-OFF CORE A (1 OF 5)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_0.5_SOA	2 1/2	8 1/2	1 1/2	1 1/2	0.5	0.8
TR_0.85_SOA	4	10	1 1/2	1 1/2	0.85	1.1
TR_1_SOA	5	15	1 1/2	1 1/2	1	1.9
TR_2_SOA	10	20	1 1/2	1 1/2	2	3.2
TR0_2.2_SOA	6	16	2	2	2.2	3.2
TR1_2.2_SOA	6 1/2	16 1/2	3	1 1/2	2.2	3.2
TR0_3_SOA	11	21	1 1/2	2	3	4.3
TR1_3_SOA	4	12	4	2	3	3.7
TR2_3_SOA	8	18	2	2	3	4.1
TR_3.8_SOA	8	18	4	1 1/2	3.8	4.8
TR_4.1_SOA	12	22	3	1 1/2	4.1	5.4
TR_5_SOA	24	38	1 1/2	1 1/2	5	7.3
TR_5.7_SOA	30	44	1 1/2	1	5.7	8.3
TR_5.8_SOA	5 1/2	15 1/2	5 1/2	2	5.8	6.7

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE A (2 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_6.3_SOA	6	16	5 1/2	2	6.3	7.3
TR_6.5_SOA	22	36	1 1/2	2	6.5	8.7
TR_7.7_SOA	9	19	4 1/2	2	7.7	8.8
TR0_8_SOA	12	22	2	3 1/2	8	9.3
TR1_8_SOA	8	18	3	3 1/2	8	9.1
TR_8.7_SOA	8	18	3	3 1/2	8.7	9.8
TR_8.8_SOA	24	38	2	2	8.8	11.1
TR_10_SOA	48	62	1 1/2	1 1/2	10	10.9
TR1_10_SOA	10	20	3 1/2	3	10	10.9
TR_10.5_SOA	9	19	3 1/2	3 1/2	10.5	11.6
TR_11_SOA	17	27	2	3 1/2	11	13
TR0_12_SOA	12	27	3 1/2	3	12	13
TR1_12_SOA	60	78	1 1/2	1 1/2	12	17
TR_13_SOA	10	20	5 1/2	2	13	14
TR_14_SOA	18	28	4	2	14	15
TR_16_SOA	23	37	3	2 1/2	16	19
TR_17_SOA	20 1/2	38 1/2	4 1/2	2	17	19
TR0_18_SOA	48	66	2	2	18	22
TR1_18_SOA	13 1/2	23 1/2	5 1/2	2 1/2	18	19
TR2_18_SOA	21 1/2	35 1/2	3 1/2	3	18	20
TR_20_SOA	30	37	2	3 1/2	20	22

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE A (3 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR0_22_SOA	20	30	4 1/2	2 1/2	22	23
TR1_22_SOA	60	78	2	2	22	27
TR_25_SOA	16	26	4	4	25	26
TR0_26_SOA	24	38	3	3 1/2	26	28
TR1_26_SOA	24	38	3 1/2	3 1/2	26	28
TR0_28_SOA	48	66	2 1/2	2 1/2	28	32
TR_32_SOA	24	38	5 1/2	2 1/2	32	34
TR_34_SOA	22	36	4	4	34	36
TR_36_SOA	60	78	2 1/2	2 1/2	36	40
TR_38_SOA	60	78	2 1/2	2 1/2	38	43
TR0_42_SOA	48	66	3	3	42	45
TR1_42_SOA	36	50	5	2 1/2	42	45
TR2_42_SOA	36	50	3 1/2	3 1/2	42	45
TR_47_SOA	55	73	3	3	47	51
TR0_48_SOA	20	30	4 1/2	5 1/2	48	50
TR1_48_SOA	20	30	5	5	48	50
TR_50_SOA	60	78	2 1/2	3 1/2	50	55
TR0_52_SOA	60	78	3	3	52	56
TR1_52_SOA	22 1/2	36 1/2	5	5	52	54
TR_58_SOA	60	74	2 1/2	4	58	60
TR_59_SOA	30	44	4 1/2	4 1/2	59	61

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE A (4 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_65_SOA	42	60	4	4	65	68
TR_72_SOA	23	37	6 1/2	5	72	75
TR_74_SOA	48	66	4	4	74	78
TR_76_SOA	44	62	4	4 1/2	76	80
TR_86_SOA	28	42	6	5 1/2	86	89
TR_87_SOA	36	50	5	5	87	90
TR_92_SOA	60	78	4	4	92	97
TR_93_SOA	48	66	5	4	93	97
TR_95_SOA	36	50	6	4 1/2	95	98
TR_97_SOA	45 1/2	63 1/2	4 1/2	5	97	101
TR_103_SOA	45	59	4 1/2	5 1/2	103	106
TR_105_SOA	60 1/2	74	4	4 1/2	105	108
TR_125_SOA	40	54	6	5 1/2	125	128
TR_130_SOA	42	56	6	5 1/2	130	133
TR_134_SOA	77	91	4 1/2	4	134	138
TR_138_SOA	40	54	6 1/2	5 1/2	138	142
TR_140_SOA	40	54	6 1/2	5 1/2	140	144
TR_148_SOA	42 1/2	56	6 1/2	5 1/2	148	152
TR_150_SOA	77	91	4 1/2	4 1/2	150	155
TR_152_SOA	40 1/2	54	6 1/2	6	152	156
TR_160_SOA	60	74	5	5 1/2	160	165

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE A (5 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR0_165_SOA	36	50	7	7	165	168
TR1_165_SOA	48	62	10 1/2	3 1/2	165	168
TR_174_SOA	48	62	10	3 1/2	174	178
TR_182_SOA	72	86	4 1/2	5 1/2	182	186
TR_185_SOA	60 1/2	74	5 1/2	6	185	189
TR_188_SOA	73	87	4 1/2	5 1/2	188	192
TR_195_SOA	61	75	5 1/2	6	195	199
TR_231_SOA	42 1/2	56	7	8	231	236
TR_243_SOA	51	65	7	7	243	248
TR_258a_SOA	51 1/2	65	7	7	244	249
TR_245_SOA	51 1/2	65	7	7	245	250
TR_258_SOA	28	42	10 1/2	9	258	263
TR_266_SOA	49	63	7	8	266	271
TR_304_SOA	48	62	6 1/2	10	304	310

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE B (1 OF 5)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_0.5_SOB	2 1/2	3 1/2	1 1/2	1 1/2	0.5	0.8
TR_0.85_SOB	4	6	1 1/2	1 1/2	0.85	1.1
TR_1_SOB	5	7	1 1/2	1 1/2	1	1.9
TR_2_SOB	10	14	1 1/2	1 1/2	2	3.2
TR0_2.2_SOB	6	8	2	2	2.2	3.2
TR1_2.2_SOB	6 1/2	8 1/2	3	1 1/2	2.2	3.2
TR0_3_SOB	11	15	1 1/2	2	3	4.3
TR1_3_SOB	4	5	4	2	3	3.7
TR2_3_SOB	8	10	2	2	3	4.1
TR_3.8_SOB	8	10	4	1 1/2	3.8	4.8
TR_4.1_SOB	12	16	3	1 1/2	4.1	5.4
TR_5_SOB	24	32	1 1/2	1 1/2	5	7.3
TR_5.7_SOB	30	36	1 1/2	1	5.7	8.3
TR_5.8_SOB	5 1/2	7 1/2	5 1/2	2	5.8	6.7

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE B (2 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_6.3_SOB	6	8	5 1/2	2	6.3	7.3
TR_6.5_SOB	22	28	1 1/2	2	6.5	8.7
TR_7.7_SOB	9	11	4 1/2	2	7.7	8.8
TR0_8_SOB	12	16	2	3 1/2	8	9.3
TR1_8_SOB	8	10	3	3 1/2	8	9.1
TR_8.8_SOB	24	30	2	2	8.8	11.1
TR0_10_SOB	10	14	3 1/2	3	10	11.2
TR1_10_SOB	48	54	1 1/2	1 1/2	10	11.2
TR_10.5_SOB	9	13	3 1/2	3 1/2	10.5	11.6
TR_11_SOB	17	21	2	3 1/2	11	13
TR0_12_SOB	12	16	3 1/2	3	12	13
TR1_12_SOB	60	66	1 1/2	1 1/2	12	17
TR_13_SOB	10	14	5 1/2	2 1/2	13	14
TR_14_SOB	18	22	4	2	14	15
TR_16_SOB	23	29	3	2 1/2	16	19
TR_17_SOB	20 1/2	26 1/2	4 1/2	2	17	19
TR0_18_SOB	48	54	2	2	18	22
TR1_18_SOB	13 1/2	17 1/2	5 1/2	2 1/2	18	19
TR2_18_SOB	21 1/2	27 1/2	3 1/2	3	18	20
TR_20_SOB	30	36	2	3 1/2	20	22
TR0_22_SOB	20	26	4 1/2	2 1/2	22	23

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE B (3 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR1_22_SOB	60	66	2	2	22	27
TR_25_SOB	16	20	4	4	25	26
TR0_26_SOB	24	30	3	3 1/2	26	28
TR1_26_SOB	24	30	3 1/2	3 1/2	26	28
TR0_28_SOB	48	54	2 1/2	2 1/2	28	32
TR1_28_SOB	48	54	2 1/2	2 1/2	28	32
TR_32_SOB	24	30	5 1/2	2 1/2	32	34
TR_34_SOB	22	28	4	4	34	36
TR_36_SOB	60	66	2 1/2	2 1/2	36	40
TR0_42_SOB	48	54	3	3	42	45
TR1_42_SOB	36	42	5	2 1/2	42	45
TR2_42_SOB	36	42	3 1/2	3 1/2	42	45
TR_47_SOB	55	61	3	3	47	51
TR0_48_SOB	20	26	4 1/2	5 1/2	48	50
TR1_48_SOB	20	26	5	5	48	50
TR_50_SOB	60	66	2 1/2	3 1/2	50	55
TR0_52_SOB	60	66	3	3	52	54
TR1_52_SOB	22 1/2	28 1/2	5	5	52	54
TR_58_SOB	60	66	2 1/2	4	58	60
TR_59_SOB	30	36	4 1/2	4 1/2	59	61
TR_65_SOB	42	48	4	4	65	68

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE B (4 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_72_SOB	23	29	6 1/2	5	72	76
TR_74_SOB	48	54	4	4	74	78
TR_76_SOB	44	50	4	4 1/2	76	80
TR_86_SOB	28	34	6	5 1/2	86	89
TR_87_SOB	36	42	5	5	87	90
TR_92_SOB	60	66	4	4	92	97
TR_93_SOB	48	54	5	4	93	97
TR_95_SOB	36	42	6	4 1/2	95	98
TR_97_SOB	45 1/2	51 1/2	4 1/2	5	97	101
TR_103_SOB	45	51	4 1/2	5 1/2	103	106
TR_105_SOB	60 1/2	66 1/2	4	4 1/2	105	108
TR_125_SOB	40	46	6	5 1/2	125	128
TR_130_SOB	42	48	6	5 1/2	130	133
TR_134_SOB	77	83	4 1/2	4	134	138
TR_138_SOB	40	46	6 1/2	5 1/2	138	142
TR_140_SOB	40	46	6 1/2	5 1/2	140	144
TR_148_SOB	42 1/2	48 1/2	6 1/2	5 1/2	148	152
TR_150_SOB	77	83	4 1/2	4 1/2	150	155
TR_152_SOB	40 1/2	46 1/2	6 1/2	6	152	156
TR_160_SOB	60	66	5	5 1/2	160	165
TR0_165_SOB	36	42	7	7	165	168

Standard sizes shown. Any weight and size required by the customer can be manufactured.



STAND-OFF CORE B (5 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR1_165_SOB	48	54	10 1/2	3 1/2	165	168
TR_174_SOB	48	54	10	3 1/2	174	178
TR_182_SOB	72	77	4 1/2	5 1/2	182	186
TR_185_SOB	60 1/2	60 1/2	5 1/2	6	185	189
TR_188_SOB	73	87	4 1/2	5 1/2	188	192
TR_195_SOB	61	67	5 1/2	6	195	199
TR_231_SOB	42 1/2	48 1/2	7	8	231	236
TR_243_SOB	51	56	7	7	243	248
TR_244_SOB	51 1/2	56 1/2	7	7	244	249
TR_245_SOB	51 1/2	56 1/2	7	7	245	250
TR_258_SOB	28	34	10 1/2	9	258	263
TR_266_SOB	49	55	7	8	266	271
TR_304_SOB	48	54	6 1/2	10	304	310



ROD CORE (1 OF 5)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_0.5_R	2 1/2	3 1/2	1 1/2	1 1/2	0.5	0.8
TR_0.85_R	4	6	1 1/2	1 1/2	0.85	1.1
TR_1_R	5	7	1 1/2	1 1/2	1	1.9
TR_2_R	10	14	1 1/2	1 1/2	2	3.2
TR0_2.2_R	6	8	2	2	2.2	3.2
TR1_2.2_R	6 1/2	8 1/2	3	1 1/2	2.2	3.2
TR0_3_R	11	15	1 1/2	2	3	4.3
TR1_3_R	4	5	4	2	3	3.7
TR2_3_R	8	10	2	2	3	4.1
TR_3.8_R	8	10	4	1 1/2	3.8	4.8
TR_4.1_R	12	16	3	1 1/2	4.1	5.4
TR_5_R	24	32	1 1/2	1 1/2	5	7.3
TR_5.7_R	30	36	1 1/2	1	5.7	8.3
TR_5.8_R	5 1/2	7 1/2	5 1/2	2	5.8	6.7

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD CORE (2 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_6.3_R	6	8	5 1/2	2	6.3	7.3
TR_6.5_R	22	28	1 1/2	2	6.5	8.7
TR_7.7_R	9	11	4 1/2	2	7.7	8.8
TR0_8_R	12	16	2	3 1/2	8	9.1
TR1_8_R	8	10	3	3 1/2	8	9.1
TR0_8.8_R	24	30	2	2	8.8	11.1
TR1_10_R	10	14	3 1/2	3	10	11.2
TR1b_10_R	48	54	1 1/2	1 1/2	10	11.2
TR_10.5_R	9	13	3 1/2	3 1/2	10.5	11.6
TR_11_R	17	21	2	3 1/2	11	13
TR0_12_R	12	16	3 1/2	3	12	13
TR1_12_R	60	66	1 1/2	1 1/2	12	17
TR_13_R	10	14	5 1/2	2 1/2	13	14
TR_14_R	18	22	4	2	14	15
TR_16_R	23	29	3	2 1/2	16	19
TR_17_R	20 1/2	26 1/2	4 1/2	2	17	19
TR0_18_R	48	54	2	2	18	22
TR1_18_R	13 1/2	17 1/2	5 1/2	2 1/2	18	19
TR2_18_R	21 1/2	27 1/2	3 1/2	3	18	20
TR_20_R	30	36	2	3 1/2	20	22
TR0_22_R	20	26	4 1/2	2 1/2	22	23

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD CORE (3 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR1_22_R	60	66	2	2	22	27
TR_25_R	16	20	4	4	25	26
TR0_26_R	24	30	3	3 1/2	26	28
TR1_26_R	24	30	3 1/2	3 1/2	26	28
TR0_28_R	48	54	2 1/2	2 1/2	28	32
TR1_28_R	48	54	2 1/2	2 1/2	28	32
TR_32_R	24	30	5 1/2	2 1/2	32	34
TR_34_R	22	28	4	4	34	36
TR_36_R	60	66	2 1/2	2 1/2	36	40
TR0_42_R	48	54	3	3	42	45
TR1_42_R	36	42	5	2 1/2	42	45
TR2_42_R	36	42	3 1/2	3 1/2	42	45
TR_47_R	55	61	3	3	47	51
TR0_48_R	20	26	4 1/2	5 1/2	48	50
TR1_48_R	20	26	5	5	48	50
TR_50_R	60	66	2 1/2	3 1/2	50	55
TR0_52_R	60	66	3	3	52	54
TR1_52_R	22 1/2	28 1/2	5	5	52	54
TR_58_R	60	66	2 1/2	4	58	60
TR_59_R	30	36	4 1/2	4 1/2	59	61
TR_65_R	42	48	4	4	65	68

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD CORE (4 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_72_R	23	29	6 1/2	5	72	76
TR_74_R	48	54	4	4	74	78
TR_76_R	44	50	4	4 1/2	76	80
TR_86_R	28	34	6	5 1/2	86	89
TR_87_R	36	42	5	5	87	90
TR_92_R	60	66	4	4	92	97
TR_93_R	48	54	5	4	93	97
TR_95_R	36	42	6	4 1/2	95	98
TR_97_R	45 1/2	51 1/2	4 1/2	5	97	101
TR_103_R	45	51	4 1/2	5 1/2	103	106
TR_105_R	60 1/2	66 1/2	4	4 1/2	105	108
TR_125_R	40	46	6	5 1/2	125	128
TR_130_R	42	48	6	5 1/2	130	133
TR_134_R	77	83	4 1/2	4	134	138
TR_138_R	40	46	6 1/2	5 1/2	138	142
TR_140_R	40	46	6 1/2	5 1/2	140	144
TR_148_R	42 1/2	48 1/2	6 1/2	5 1/2	148	152
TR_150_R	77	83	4 1/2	4 1/2	150	155
TR_152_R	40 1/2	46 1/2	6 1/2	6	152	156
TR_160_R	60	66	5	5 1/2	160	165
TR0_165_R	36	42	7	7	165	168

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD CORE (5 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR1_165_R	48	54	10 1/2	3 1/2	165	168
TR_174_R	48	54	10	3 1/2	174	178
TR_182_R	72	77	4 1/2	5 1/2	182	186
TR_185_R	60 1/2	66 1/2	5 1/2	6	185	189
TR_188_R	73	87	4 1/2	5 1/2	188	192
TR_195_R	61	67	5 1/2	6	195	199
TR_231_R	42 1/2	48 1/2	7	8	231	236
TR_243_R	51	56	7	7	243	248
TR_258a_R	51 1/2	56 1/2	7	7	244	249
TR_245_R	51 1/2	56 1/2	7	7	245	250
TR_258_R	28	34	10 1/2	9	258	263
TR_266_R	49	55	7	8	266	271
TR_304_R	48	54	6 1/2	10	304	310

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD STAND-OFF CORE (1 OF 5)



NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_0.5_RSO	2 1/2	8 1/2	1 1/2	1 1/2	0.5	0.8
TR_0.85_RSO	4	14	1 1/2	1 1/2	0.85	1.1
TR_1_RSO	5	15	1 1/2	1 1/2	1	1.9
TR_2_RSO	10	20	1 1/2	1 1/2	2	3.2
TR0_2.2_RSO	6	16	2	2	2.2	3.2
TR1_2.2_RSO	6 1/2	16 1/2	3	1 1/2	2.2	3.2
TR0_3_RSO	11	21	1 1/2	2	3	4.3
TR1_3_RSO	4	12	4	2	3	3.7
TR2_3_RSO	8	18	2	2	3	4.1
TR_3.8_RSO	8	18	4	1 1/2	3.8	4.8
TR_4.1_RSO	12	22	3	1 1/2	4.1	5.4
TR_5_RSO	24	38	1 1/2	1 1/2	5	7.3
TR_5.7_RSO	30	44	1 1/2	1	5.7	8.3
TR_5.8_RSO	5 1/2	15 1/2	5 1/2	2	5.8	6.7

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD STAND-OFF CORE (2 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_6.3_RSO	6	16	5 1/2	2	6.3	7.3
TR_6.5_RSO	22	36	1 1/2	2	6.5	8.7
TR_7.7_RSO	9	19	4 1/2	2	7.7	8.8
TR0_8_RSO	12	22	2	3 1/2	8	9.3
TR1_8_RSO	8	18	3	3 1/2	8	9.1
TR_8.8_RSO	24	38	2	2	8.8	11.1
TR0_10_RSO	10	20	3 1/2	3	10	11.2
TR1_10_RSO	48	62	1 1/2	1 1/2	10	11.2
TR_10.5_RSO	9	19	3 1/2	3 1/2	10.5	11.6
TR_11_RSO	17	27	2	3 1/2	11	13
TR0_12_RSO	12	22	3 1/2	3	12	13
TR1_12_RSO	60	78	1 1/2	1 1/2	12	17
TR_13_RSO	10	20	5 1/2	2 1/2	13	14
TR_14_RSO	18	28	4	2	14	15
TR_16_RSO	23	37	3	2 1/2	16	19
TR_17_RSO	20 1/2	38 1/2	4 1/2	2	17	19
TR0_18_RSO	48	66	2	2	18	22
TR1_18_RSO	13 1/2	23 1/2	5 1/2	2 1/2	18	19
TR2_18_RSO	21 1/2	35 1/2	3 1/2	3	18	20
TR_20_RSO	30	37	2	3 1/2	20	22
TR0_22_RSO	20	30	4 1/2	2 1/2	22	23

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD STAND-OFF CORE (3 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR1_22_RSO	60	78	2	2	22	27
TR_25_RSO	16	26	4	4	25	26
TR0_26_RSO	24	38	3	3 1/2	26	28
TR1_26_RSO	24	38	3 1/2	3 1/2	26	28
TR0_28_RSO	48	66	2 1/2	2 1/2	28	32
TR_32_RSO	24	38	5 1/2	2 1/2	32	34
TR_34_RSO	22	36	4	4	34	36
TR_36_RSO	60	78	2 1/2	2 1/2	36	40
TR0_42_RSO	48	66	3	3	42	45
TR1_42_RSO	36	43	5	2 1/2	42	45
TR2_42_RSO	36	43	3 1/2	3 1/2	42	45
TR_47_RSO	55	73	3	3	47	51
TR0_48_RSO	20	30	4 1/2	5 1/2	48	50
TR1_48_RSO	20	30	5	5	48	50
TR_50_RSO	60	78	2 1/2	3 1/2	50	55
TR0_52_RSO	60	78	3	3	52	54
TR1_52_RSO	22 1/2	36 1/2	5	5	52	54
TR_58_RSO	60	64	2 1/2	4	58	60
TR_59_RSO	30	44	4 1/2	4 1/2	59	61
TR_65_RSO	42	60	4	4	65	68
TR_72_RSO	23	37	6 1/2	5	72	76

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD STAND-OFF CORE (4 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_74_RSO	48	66	4	4	74	78
TR_76_RSO	44	62	4	4 1/2	76	80
TR_86_RSO	28	42	6	5 1/2	86	89
TR_87_RSO	36	50	5	5	87	90
TR_92_RSO	60	78	4	4	92	97
TR_93_RSO	48	66	5	4	93	97
TR_95_RSO	36	50	6	4 1/2	95	98
TR_97_RSO	45 1/2	63 1/2	4 1/2	5	97	101
TR_103_RSO	45	59	4 1/2	5 1/2	103	106
TR_105_RSO	60 1/2	74	4	4 1/2	105	108
TR_125_RSO	40	54	6	5 1/2	125	128
TR_130_RSO	42	56	6	5 1/2	130	133
TR_134_RSO	77	91	4 1/2	4	134	138
TR_138_RSO	40	54	6 1/2	5 1/2	138	142
TR_140_RSO	40	54	6 1/2	5 1/2	140	144
TR_148_RSO	42 1/2	56	6 1/2	5 1/2	148	152
TR_150_RSO	77	91	4 1/2	4 1/2	150	155
TR_152_RSO	40 1/2	54	6 1/2	6	152	156
TR_160_RSO	60	74	5	5 1/2	160	165
TR0_165_RSO	36	50	7	7	165	168
TR1_165_RSO	48	62	10 1/2	3 1/2	165	168

Standard sizes shown. Any weight and size required by the customer can be manufactured.



ROD STAND-OFF CORE (5 OF 5)

NAME	La (in)	Lc (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)
TR_174_RSO	48	62	10	3 1/2	174	178
TR_182_RSO	72	86	4 1/2	5 1/2	182	186
TR_185_RSO	60 1/2	74	5 1/2	6	185	189
TR_188_RSO	73	87	4 1/2	5 1/2	188	192
TR_195_RSO	61	75	5 1/2	6	195	199

HOUSTON ANODES

TANKS AND VESSELS ANODES

Houston Anodes manufactures aluminum and zinc sacrificial anodes engineered specifically for internal vessel applications, protecting against corrosion in the harshest conditions. Vessel and tank anodes are cast in one of two premium aluminum alloys.

Houston Anodes can cast anodes for any tank bottom or vessel interior to the exact specifications required. Contact our engineering team for custom anode optimization tailored to your specific vessel requirements.

01 PRODUCT TYPES

Heater Treater

Heater Treater Eyebolt

Bowling Ball

Drum

ALUMINUM ALLOY

ZINC ALLOY

CUSTOM CROSS-SECTIONS





HEATER TREATER



NAME	La (in)	Ø (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
HT_18.5_1R	30	3	18.5	23	¾" ROD
HT_27.75_1R	45	3	27.75	34	¾" ROD
HT_37_1R	60	3	37	45	¾" ROD

HOUSTON ANODES



HEATER TREATER EYEBOLT



NAME	La (in)	Ø (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
HT_18.5_1RR	30	3	18.5	23	¾" ROD
HT_27.75_1RR	45	3	27.75	34	¾" ROD
HT_37_1RR	60	3	37	45	¾" ROD

HOUSTON ANODES



BOWLING BALL



NAME	La (in)	Ø (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
BB_46	4	9.8	46	47	1" SCH. 80

HOUSTON ANODES



DRUM



NAME	La (in)	Ø (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
DR_45	9 ¼	8 ¼	45	46	1" SCH. 80

HOUSTON ANODES

ZINC HULLS

Zinc hull anodes for marine equipment and vessel protection. Available in multiple configurations for new construction and retrofit applications. U.S. Navy anodes with proven 35-year service life.

01 CONFIGURATIONS

Trapezoidal Hull

Trapezoidal Hull — Cast-in Core

Hull Tapered

Tapered Trapezoidal

Teardrop

ALUMINUM ALLOY

ZINC ALLOY

CUSTOM CROSS-SECTIONS

HOUSTON ANODES





TRAPEZOIDAL HULL



NAME	L (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
H_23_2S	12	6	1 ¼	23	24	3/16 x ¼"
H_37_2S	12	6	2	37	38	3/16 x ¼"
H_39_2S	17 ½	4 ½	2	39	40	3/16 x ¼"
H_46_2S	12	6	2 ½	46	47	3/16 x ¼"
H_85_2S	18	9 ½	2	85	87	¼ x 2"

HOUSTON ANODES



TRAPEZOIDAL HULL – CAST-IN CORE



NAME	L (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
H_21_2S	12	6	1 ¼	21	22	3/16 x ¼"
H_31_2S	12	6	2	31	32	3/16 x ¼"
H_38_2S	17 ¼	4 ½	2	38	39	3/16 x ¼"
H_41_2S	12	6	2 ½	41	42	3/16 x ¼"
H_82_2S	18	9 ½	2	82	84	3/16 x ¼"

HOUSTON ANODES



HULL TAPERED



NAME	L (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
TH_22.5_2S	14	6 ½	1 ¼	22.5	23.5	

HOUSTON ANODES



TAPERED TRAPEZOIDAL



NAME	L (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
TT_6_1S	6	2 ½	1 ¼	6	6	¼x1 ½
TT_11_1S	8	4	1 ¼	11	12	¼x1 ½
TT_37_1S	12	6	2	37	39	¼x1 ½
TT_63_1S	21	5	2 ½	63	65	¼x1 ½
TT_73_1S	24	5 ½	2 ½	73	75	¼x1 ½

HOUSTON ANODES



TEARDROP



NAME	L (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
TD_3_1S	5 ¼	2 ½	1 ¼	3	3	1/8"x3/4"
TT_15_1S	9	3 ½	2	15	15	1/8"x3/4"
TT_37b_1S	11	5 ½	2 ½	37	38	1/8"x3/4"
TT_67_1S	12 ½	6 ½	3 ½	67	68	1/8"x3/4"

HOUSTON ANODES

CUSTOM & REFERENCE ANODES

Houston Anodes can fabricate any type, any size, and any shape of anode to accommodate customer-specific needs. The company manufactures its own molds for complete production control. Available in aluminum and zinc alloys.

Custom weights from 5 lbs to 1,650 lbs in custom configurations. Any geometry manufactured to exact specifications, including anodes for offshore wind foundations.

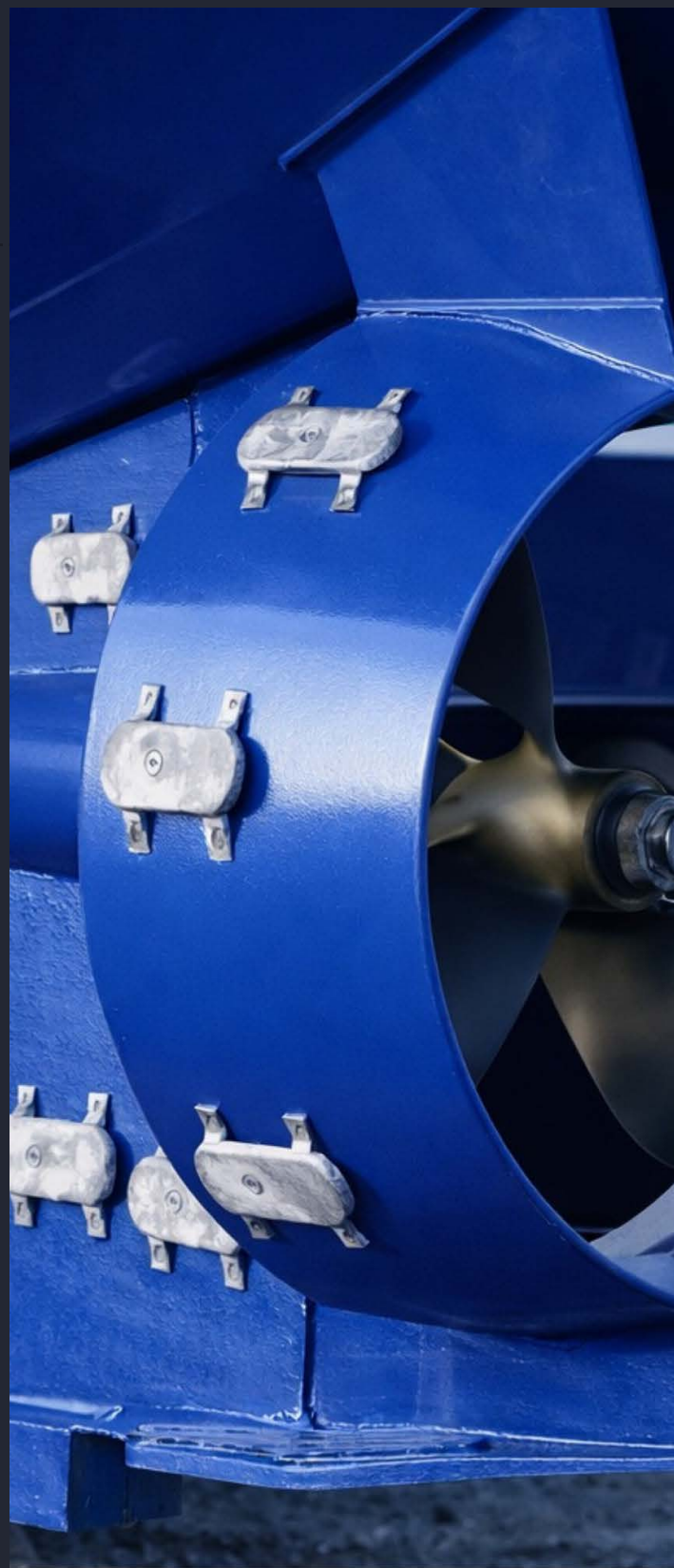
The world's biggest anode! BP Juniper Project, 2015 — by Houston Anodes.

ALUMINUM ALLOY

ZINC ALLOY

CUSTOM CROSS-SECTIONS

HOUSTON ANODES





CLAMP



NAME	L1 (in)	L2 (in)	W (in)	H (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
CL_16	11 ½	5	4	4 ½	16	16	½" SCH. 80

HOUSTON ANODES



MAT



NAME	L (in)	L2 (in)	Net Weight (lbs)	Gross Weight (lbs)
SM_17	4 ½	7 3/8	17	18

HOUSTON ANODES



LINK (1 UNIT)



NAME	L (in)	L2 (in)	Net Weight (lbs)	Gross Weight (lbs)	Core Type
RL_45.5	60	4 ½	45.5	50	¼" STEEL CABLE

HOUSTON ANODES



CERTIFICATIONS & STANDARDS

Houston Anodes manufacturing and laboratory operations are independently accredited and audited against the standards governing cathodic protection materials worldwide.

ISO 9001:2015

Certified by DNV

ISO/IEC 17025

Accredited laboratory

NACE TM-0190

Electrochemical test method

QUALITY YOU CAN TRACE

All raw materials, finished anodes, and chemical and electrochemical test data are traceable to Mill Test Reports and retained per Houston Anodes' quality management system. Certificates of Conformance are issued with every shipment.

EVERY SHIPMENT

Certificate of Conformance
issued

EVERY HEAT

Chemical analysis performed

EVERY BATCH

Electrochemical testing per
NACE TM-0190

FULL TRACEABILITY

Mill Test Reports retained

HOUSTON ANODES



CONTACT INFORMATION

Houston, TX · Energy Corridor

832.243.0700

anodes@houstonanodes.com

[FACEBOOK](#) · [LINKEDIN](#) · [X](#)

THANK YOU FOR CHOOSING HOUSTON ANODES